Alina Zaman

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Research Interest

My research interests lie in **the security and privacy of machine learning systems and software supply chains**, centered on what happens when systems depend on external code or data they cannot fully verify. Professionally, I build tools for CVE reachability analysis through dependency chains and call graphs, and I am currently researching the security and robustness of federated learning. My undergraduate work on convolutional feature aggregation for image retrieval was published at IEEE ICMI 2026.

Education

Bangladesh University of Engineering and Technology Feb 2020 – March 2025
B.Sc. in Computer Science and Engineering ***CGPA : 3.55/4.00 [Senior Year : 3.96/4.00]***
Relevant Coursework: Machine Learning, Computer Security, Computer Networks, Software Engineering, Algorithm Engineering, Basic Graph Theory **(all completed with a 4.00/4.00)**

Shaheed Bir Uttam LT Anwar Girls' College July 2017 – April 2019
Higher Secondary Certificate (HSC) ***GPA : 5.00/5.00***

Agargaon Taltola Govt. Colony High School and College Jan 2015 – Feb 2017
Secondary School Certificate (SSC) ***GPA : 5.00/5.00***

Publications

[1] Extending Feature Selection Strategies in VGG16: Convolutional Feature Aggregation for Content-Based Image Retrieval, 2026

Saha Kuljit Shantanu, ***Alina Zaman****, Md Monirul Islam*

Published at 2026 IEEE 5th International Conference on Computing and Machine Intelligence (ICMI) | Paper Link

Research Experience

Adversarial Machine Learning in Federated Handover Prediction

Ongoing Research Project
Supervisors: Assoc. Prof. Sukarna Barua, Dept. of CSE, BUET | Assoc. Prof. Nashid Shahriar, Dept. of CS, University of Regina

- Investigating security vulnerabilities of federated learning models used for handover prediction in 5G networks, including model poisoning, backdoor, and evasion attacks.
- Defining threat models and an experimental framework for evaluating attack impact on network-level metrics.

Attacks and Defenses in Federated Learning April 2025 – June 2025

Research Collaboration
Mentor: Md Shahedul Haque, then MS student, Virginia Tech

- Following a literature review and paper presentation on attacks and defenses in federated learning, reproduced Bhagoji et al., "Analyzing Federated Learning through an Adversarial Lens" (ICML 2019), on a 10-client federated system built on Adult Census with an MLP implemented from scratch, covering its three model-poisoning attacks (explicit boosting, stealthy poisoning, alternating minimization).
- Across 60 runs (6 attack/defense setups × 5 seeds × 2 target directions), matched the paper's FedAvg results and found that Krum and coordinate-wise median blocked the attacks tested against them in every valid run. [Code Link]

Adaptive Feature Selection for Content-Based Image Retrieval (CBIR) Oct 2023 – March 2025

Undergraduate Thesis (Grade: 4.00/4.00)
Supervisor: Prof. Md. Monirul Islam, Dept. of CSE, BUET

- Co-developed an adaptive multi-channel aggregation method for CNN-based image retrieval, replacing the single-dominant-channel selection used by prior work with a dataset-driven threshold that fuses multiple high-ranking convolutional channels.
- Implemented the full retrieval pipeline over frozen VGG16 conv5_3 features: dominance-ranked channel fusion, spatial masking, Gaussian spatial weighting, and PCA-whitening to 100-d descriptors.
- Improved mAP by 2.3 points over the AdCoW+I baseline on both Oxford5K and Paris6K, with gains consistent across all tested threshold values.
- Analyzed the effect of monochromatic input on retrieval, identifying a dataset-dependent effect that motivates follow-up investigation.

Professional Experience

OpenRefactory Dhaka, Bangladesh
Software Engineer July 2025 – Present, Full-time

- Contributed to the algorithm design and implementation of a VEX generation tool that analyzes CVE reachability using root cause information, dependency chains, and call graphs for both open-source and proprietary code.
- Built a license scanning tool from scratch in Python that detects licenses through static text analysis of license files, copyright headers of source code files, and package metadata, with LLM-based classification as a fallback when static analysis is inconclusive. Output is used to analyze licenses across dependency trees and flag policy violations for open-source license compliance.
- Contributed to the backend and microservices of ProjectClean, including efficient dependency tree generation for artifacts, query optimization for UI performance, and API development.

Skills

Programming: Python, C/C++, Java, JavaScript, SQL

Machine Learning: PyTorch, TensorFlow, Keras, OpenCV

Backend/Web: Node.js, Spring Boot, React, Next.js

Databases: PostgreSQL, OracleDB

Tools & Systems: Git, GitHub, Bash, Linux, Docker

Concepts: OOP, Functional Programming, DevOps

Academic Projects

Machine Learning Projects | *Code Link* Python, Numpy, Scikit-learn

- **Core ML Algorithms** — Implemented a complete ML pipeline combining data preprocessing, feature engineering, logistic regression with bagging/stacking, PCA, and EM clustering for pattern discovery and model evaluation.
- **Multi-Layer Perceptron (MLP)** — Implemented a feed-forward neural network from scratch, incorporating dense layers, batch normalization, dropout regularization, and the Adam optimizer to improve convergence and generalization during training.

Computer Security & Cybersecurity Projects | *Code Link* Python, Docker, Wireshark, Azure

- **Cryptography & Key Exchange** — Implemented AES encryption/decryption and elliptic-curve key generation in Python; simulated secure socket-based communication.

- **Web Security & Browser Exploitation** — In a controlled lab environment (SEED Labs), demonstrated CSRF and XSS attacks against a local test application.
- **Malware Signature Analysis with YARA** — Created and organized YARA rules for ELF, PE, console, and test-based detection patterns for static malware analysis and signature matching.

Computer Networking Projects | *Code Link* Java, Cisco Packet Tracer, NS3

- **Client-Server Networking & Reliable Data Transfer** — Implemented Java socket-based applications supporting multi-client communication, threaded servers, FTP-style file transfer, and reliable data transmission using CRC and Hamming code.
- **Network Simulation with ns-3** — Implemented an ns-3 based wireless network simulation to evaluate TCP performance under varying traffic, topology, and mobility conditions through throughput and packet delivery analysis

Computer Graphics Projects | *Code Link* Graphics, OpenGL

- **OpenGL Rendering & Animation** — Built 3D scenes with primitives, transformations, camera setup, and animated motion, including rolling/board-like scenes and cube/cylindrical object constructions.
- **3D Transformation Pipeline** — Implemented translation, rotation, scaling, and projection transformations; processed triangle data through multiple rendering stages and performed z-buffer visibility checks.
- **Ray Tracing & Shading** — Built a renderer that shoots rays from the camera through pixels, intersects them with spheres and triangles, applies ambient/diffuse/specular lighting, and exports the final images as BMP files.

C Compiler | *Code Link* Flex, Bison, YACC, C++

- Built a mini-compiler for a C-like language, covering lexical analysis, parsing, symbol table and scope management, and basic assembly-style code generation for expressions, control flow, and functions.

Thoughts | *Code(Frontend) Code(Backend)* React.js, Spring Boot

- Developed a web application integrating Spring Boot, STOMP, and LLaMA 3.2 for real-time communication and LLM-powered assistance, inspired by platforms like *The Help Line*.

Other Experiences

Bangladeshi Women in Computer Science and Engineering (BWCSE) 2023 – 2025

Served as the ***Batch Representative*** for two consecutive years.

Student Tutor 2020 – 2025

Provided academic tutoring and mentorship to students from middle school through undergraduate level, including guidance and preparation for university admissions.

References

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